# Hao-Lin Ye

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## Education

**University of Toronto** Starting September 2027
*Incoming Thesis Master student, Computer Science*

Supervisor: Prof. Xujie Si. Research Direction: agentic kernel generation and optimization, ML-assisted code verification, and deep learning interpretability.

**McGill University** September 2021 – January 2027
*B.Sc., Mathematics and Computer Science GPA: 3.78/4.00*

## Publications and Preprint

**Neural Proposals, Symbolic Guarantees: Neuro-Symbolic Graph Generative Modeling**
Chuqin Geng, Li Zhang, Mark Zhang, Zhaoyue Wang, **Haolin Ye**, Xujie Si. *NeurIPS 2026, accepted*.

**LogicXGNN: Grounded Logical Rules for Explaining Graph Neural Networks**
Chuqin Geng, Ziyu Zhao, Zhaoyue Wang, **Haolin Ye**, Yuhe Jiang, Xujie Si. *ICLR 2026*.

**Learning Minimal Neural Specifications**
Chuqin Geng, Zhaoyue Wang, **Haolin Ye**, Xujie Si. *NeuS 2025 (Oral)*.

**SAT-DIFF: A Tree Diffing Framework Using SAT Solving**
Chuqin Geng, **Haolin Ye**, Yihan Zhang, Brigitte Pientka, Xujie Si. *arXiv preprint, 2024*.

**Scalar Invariant Networks with Zero Bias**
Chuqin Geng, Xiaojie Xu, **Haolin Ye**, Xujie Si. *NeurIPS 2023 Workshop on Symmetry and Geometry in Neural Representations; PMLR 2024*.

**Novice Type Error Diagnosis with Natural Language Models**
Chuqin Geng, **Haolin Ye**, Yixuan Li, Tianyu Han, Brigitte Pientka, Xujie Si. *APLAS 2022*.

## Research Collaborations

**Prof. Xujie Si's Lab** ICLR / NeurIPS 2026 projects

*LogicXGNN*

- Contributed to the design and testing of graph-hashing-based predicate construction using local subgraph structure and GNN activation patterns.
- Evaluated predicate extraction and hyperparameter choices; ran data-grounded fidelity comparisons against GLGExplainer and GraphTrail to assess the reliability of rules and subgraph explanations.

*Neural Proposals, Symbolic Guarantees*

- Tested molecular tile decomposition, ran small-scale experiments, and contributed to proof writing for a framework combining neural proposals with SMT-based graph assembly.

## Industry Experience

**Qualcomm** May 2025 – August 2026
*Machine Learning System Engineer Intern* *Toronto, Canada*

- Applied graph optimization for AI model compilation targeting Snapdragon hardware.
- Maintained machine learning infrastructure and automated testing pipelines supporting model deployment.

## Earlier Research Experience

**McGill University** May 2023 – August 2023
*Undergraduate Research Assistant* *Montréal, Canada*
*SAT-DIFF: Tree differencing with MaxSAT*

- Developed a MaxSAT-based framework for computing minimal edit scripts between ordered trees and proposed the MaxSAT encoding of the tree-differencing problem.
- Proposed two solver optimizations and evaluated SAT-DIFF against GumTree on two benchmarks.

**Mila – Quebec AI Institute** May 2022 – May 2023
*Student Researcher* *Montréal, Canada*
*Language models for type-error diagnosis*

- Prepared datasets and performed continued pre-training and fine-tuning of BERT models for type-error localization in OCaml programs.
- Used structural probes to examine syntactic structure captured by the models.

*Neural network verification and robustness*

- Proposed a metric to estimate expansion of the verifiable input space under reduced neural specifications; applied Grad-CAM to analyze the resulting specifications.
- Evaluated zero-bias networks on computer vision benchmarks and contributed to proofs of their interpolation and convex-region robustness properties.

## Selected Project

**Assistive Cyber Eyes** June 2026
*Team Lead* *2025 Snapdragon AI Innovation Challenge, China*

- Led the team and developed an AI navigation-assistance prototype for blind users on Qualcomm SM8850 hardware.
- Won the Merit Award winner as one of 27 finalist teams from 1,000+ registered teams.

## Technical Skills

**Languages:** Python, C++, C, Java, OCaml, SQL, R.
**ML tools:** PyTorch, ONNX, NumPy, pandas, TensorFlow.
**Research methods:** SAT/MaxSAT encodings, graph algorithms, predicate extraction, neural network verification, experimental evaluation.

## Honors and Awards

**SPLASH Student Research Competition** – Second Place 2023
**McGill Undergraduate Student Research Award** 2023
**ENG WEEK Hackathon** – First Place 2022